UNIVERSIDAD MAYOR DE SAN ANDRES

PRACTICA 02

CARRERA: INFORMATICA

UNIVERSITARIA: MELANY BELEN YUJRA BAUTISTA

MATERIA: PROGRAMACION II

DOCENTE: JHONNY ROBERTO FELIPEZ ANDRADE

PARALELO: D

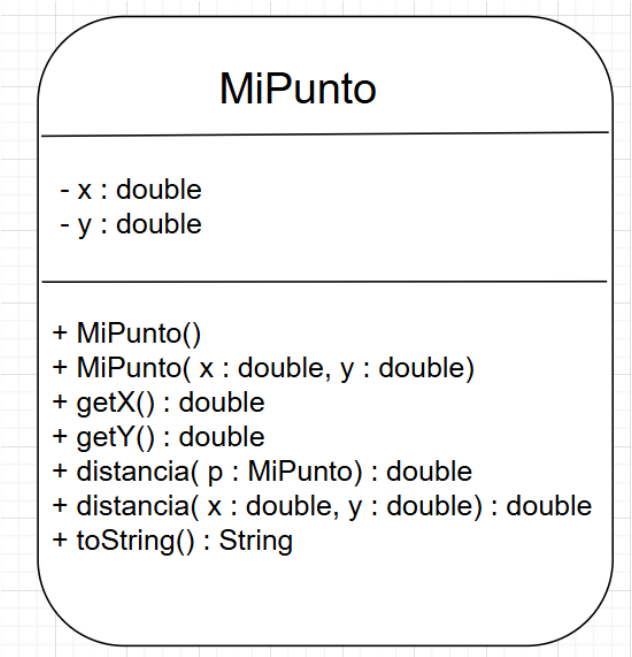
FECHA: 01 DE ABRIL DE 2026

LA PAZ - BOLIVIA

PRACTICA 1

EJERCICIO 1. MI PUNTO

DIAGRAMA UML



CODIGO EN JAVA

MiPunto.java

```
package Practica02.ejercicio1;

public class MiPunto {

    private double x;

    private double y;


    public MiPunto() {

        this.x = 0;

        this.y = 0;

    }


    public MiPunto(double x, double y) {

        this.x = x;
```

```
        this.y = y;
    }

    public double getX() {
        return x;
    }

    public double getY() {
        return y;
    }

    public double distancia(MiPunto otro) {
        double dx = this.x - otro.x;
        double dy = this.y - otro.y;
        return Math.sqrt(dx * dx + dy * dy);
    }

    public double distancia(double x, double y) {
        double dx = this.x - x;
        double dy = this.y - y;
        return Math.sqrt(dx * dx + dy * dy);
    }

    @Override
    public String toString() {
        return "(" + x + ", " + y + ")";
    }
}
```

Main.java

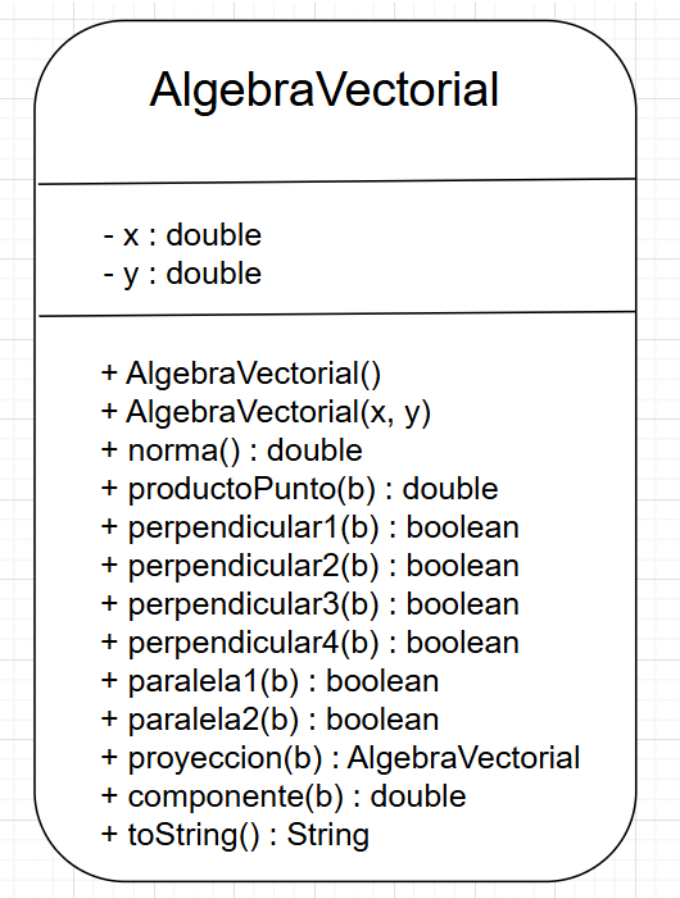
```
package Practica02.ejercicio1;
public class Main {
    public static void main(String[] args) {

        MiPunto p1 = new MiPunto();
        MiPunto p2 = new MiPunto(10, 30.5);

        System.out.println("Punto 1: " + p1);
        System.out.println("Punto 2: " + p2);

        double d1 = p1.distancia(p2);
        System.out.println("Distancia entre puntos: " + d1);

        double d2 = p1.distancia(10, 30.5);
        System.out.println("Distancia usando coordenadas: " + d2);
    }
}
```

EJECUCION

```
PROBLEMAS    SALIDA    CONSOLA DE DEPURACIÓN    TERMINAL    PUERTOS

PS C:\Users\BELEN\Desktop\Programacion II>  & 'C:\Program Files\Eclipse Adopt
ot\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages'
AppData\Roaming\Code\User\workspaceStorage\3768406a5699df1a61ce30fd7f131e14\re
amacion II_650d094f\bin' 'Practica02.ejercicio1.Main'
Punto 1: (0.0, 0.0)
Punto 2: (10.0, 30.5)
Distancia entre puntos: 32.09750769140807
Distancia usando coordenadas: 32.09750769140807
PS C:\Users\BELEN\Desktop\Programacion II>
```

EJERCICIO 2: Algebra Vectorial

DIAGRAMA UML



CODIGO EN JAVA

AlgebraVectorial.java

```
package Practica02.ejercicio2;

import static java.lang.Math.*;

public class AlgebraVectorial {

    private double x;

    private double y;

    public AlgebraVectorial() {

        this.x = 0;

        this.y = 0;

    }
```

```
public AlgebraVectorial(double x, double y) {
    this.x = x;
    this.y = y;
}
public double norma() {
    return sqrt(x * x + y * y);
}
public double productoPunto(AlgebraVectorial b) {
    return this.x * b.x + this.y * b.y;
}
public boolean perpendicular1(AlgebraVectorial b) {
    AlgebraVectorial suma = new AlgebraVectorial(x + b.x, y + b.y);
    AlgebraVectorial resta = new AlgebraVectorial(x - b.x, y - b.y);
    return abs(suma.norma() - resta.norma()) < 1e-6;
}
public boolean perpendicular2(AlgebraVectorial b) {
    AlgebraVectorial ab = new AlgebraVectorial(x - b.x, y - b.y);
    AlgebraVectorial ba = new AlgebraVectorial(b.x - x, b.y - y);
    return abs(ab.norma() - ba.norma()) < 1e-6;
}
public boolean perpendicular3(AlgebraVectorial b) {
    return abs(productoPunto(b)) < 1e-6;
}
public boolean perpendicular4(AlgebraVectorial b) {
    AlgebraVectorial suma = new AlgebraVectorial(x + b.x, y + b.y);
    return abs(pow(suma.norma(), 2) - (pow(this.norma(), 2) + pow(b.norma(), 2)))
    < 1e-6;
}
public boolean paralela1(AlgebraVectorial b) {
    if (b.x != 0) {
        double r = this.x / b.x;
```

```
            return abs(this.y - r * b.y) < 1e-6;
        } else if (b.y != 0) {
            double r = this.y / b.y;
            return abs(this.x - r * b.x) < 1e-6;
        }

        return false;
    }
    public boolean paralela2(AlgebraVectorial b) {
        return abs(this.x * b.y - this.y * b.x) < 1e-6;
    }
    public AlgebraVectorial proyeccion(AlgebraVectorial b) {
        double normaB = b.norma();
        if (normaB == 0) return new AlgebraVectorial(0, 0);

        double escalar = productoPunto(b) / (normaB * normaB);
        return new AlgebraVectorial(escalar * b.x, escalar * b.y);
    }
    public double componente(AlgebraVectorial b) {
        double normaB = b.norma();
        if (normaB == 0) return 0;

        return productoPunto(b) / normaB;
    }
    @Override
    public String toString() {
        return "(" + x + ", " + y + ")";
    }
}
```

MaIn.java

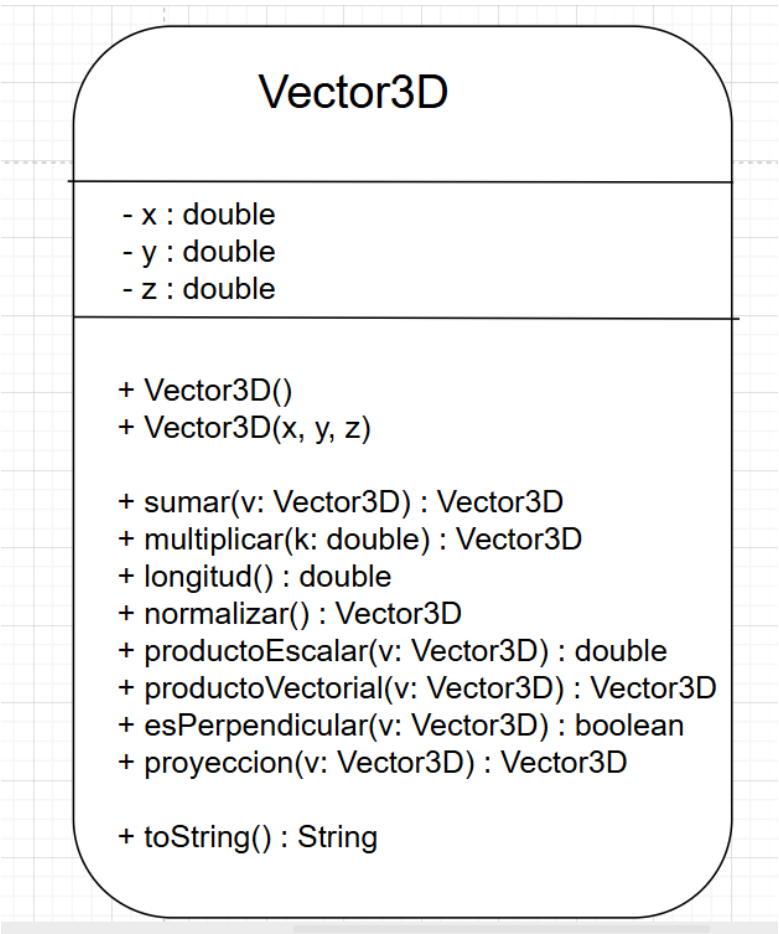
```
package Practica02.ejercicio2;
public class Main {
    public static void main(String[] args) {

        AlgebraVectorial a = new AlgebraVectorial(2, 4);
        AlgebraVectorial b = new AlgebraVectorial(1, 2);

        System.out.println("Perpendicular (a·b=0): " + a.perpendicular3(b));
        System.out.println("Paralela (a×b=0): " + a.paralela2(b));

        System.out.println("Perpendicular1: " + a.perpendicular1(b));
        System.out.println("Perpendicular2: " + a.perpendicular2(b));
        System.out.println("Perpendicular4: " + a.perpendicular4(b));

        System.out.println("Paralela1: " + a.paralela1(b));

        AlgebraVectorial proj = a.proyeccion(b);
        System.out.println("Proyección de a sobre b: " + proj);
        double comp = a.componente(b);
        System.out.println("Componente de a en b: " + comp);
    }
}
```

EJECUCION

```
5699df1a61ce30fd7f131e14\redhat.java\jdt_ws\Programacion II_650d094f\b
n'
Perpendicular (a·b=0): false
Paralela (a×b=0): true
Perpendicular1: false
Perpendicular2: true
Perpendicular4: false
Paralela1: true
Proyección de a sobre b: (1.9999999999999996, 3.999999999999999)
Componente de a en b: 4.47213595499958
PS C:\Users\BELEN\Desktop\Programacion II>
```

EJERCICIO 3: Vector 3D

DIAGRAMA UML



CODIGO EN JAVA

Vector3D.java

```
package Practica02.ejercicio3;
class Vector3D {
    double x, y, z;

    public Vector3D(double x, double y, double z) {
        this.x = x;
        this.y = y;
        this.z = z;
    }

    public Vector3D sumar(Vector3D v) {
        return new Vector3D(this.x + v.x, this.y + v.y, this.z + v.z);
    }

    public Vector3D multiplicar(double k) {
        return new Vector3D(this.x * k, this.y * k, this.z * k);
    }

    public double longitud() {
        return Math.sqrt(x*x + y*y + z*z);
    }

    public Vector3D normalizar() {
        double len = longitud();
        return new Vector3D(x/len, y/len, z/len);
    }

    public double productoEscalar(Vector3D v) {
```

```
        return this.x*v.x + this.y*v.y + this.z*v.z;
    }

    public Vector3D productoVectorial(Vector3D v) {
        double i = this.y*v.z - this.z*v.y;
        double j = this.z*v.x - this.x*v.z;
        double k = this.x*v.y - this.y*v.x;
        return new Vector3D(i, j, k);
    }

    public boolean esPerpendicular(Vector3D v) {
        return this.productoEscalar(v) == 0;
    }

    public Vector3D proyeccion(Vector3D v) {
        double escalar = this.productoEscalar(v) / v.productoEscalar(v);
        return v.multiplicar(escalar);
    }

    public String toString() {
        return "(" + x + ", " + y + ", " + z + ")";
    }
}
```

Main.java

```
package Practica02.ejercicio3;
public class Main {
    public static void main(String[] args) {

        Vector3D a = new Vector3D(1, 2, 3);
```

```
        Vector3D b = new Vector3D(4, 5, 6);

        System.out.println("Suma: " + a.sumar(b));
        System.out.println("Escalar: " + a.multiplicar(2));
        System.out.println("Longitud: " + a.longitud());
        System.out.println("Normal: " + a.normalizar());
        System.out.println("Producto escalar: " + a.productoEscalar(b));
        System.out.println("Producto vectorial: " + a.productoVectorial(b));
        System.out.println("Perpendicular: " + a.esPerpendicular(b));
        System.out.println("Proyección de a sobre b: " + a.proyeccion(b));
    }
}
```

COMPILACION

```
5699df1a61ce30fd7f131e14\redhat.java\jdt_ws\Programacion II_650d094f\bin' 'Practica02.ej
n'
Suma: (5.0, 7.0, 9.0)
Escalar: (2.0, 4.0, 6.0)
Longitud: 3.7416573867739413
Normal: (0.2672612419124244, 0.5345224838248488, 0.8017837257372732)
Producto escalar: 32.0
Producto vectorial: (-3.0, 6.0, -3.0)
Perpendicular: false
Proyección de a sobre b: (1.6623376623376624, 2.077922077922078, 2.4935064935064934)
PS C:\Users\BELEN\Desktop\Programacion II>
```